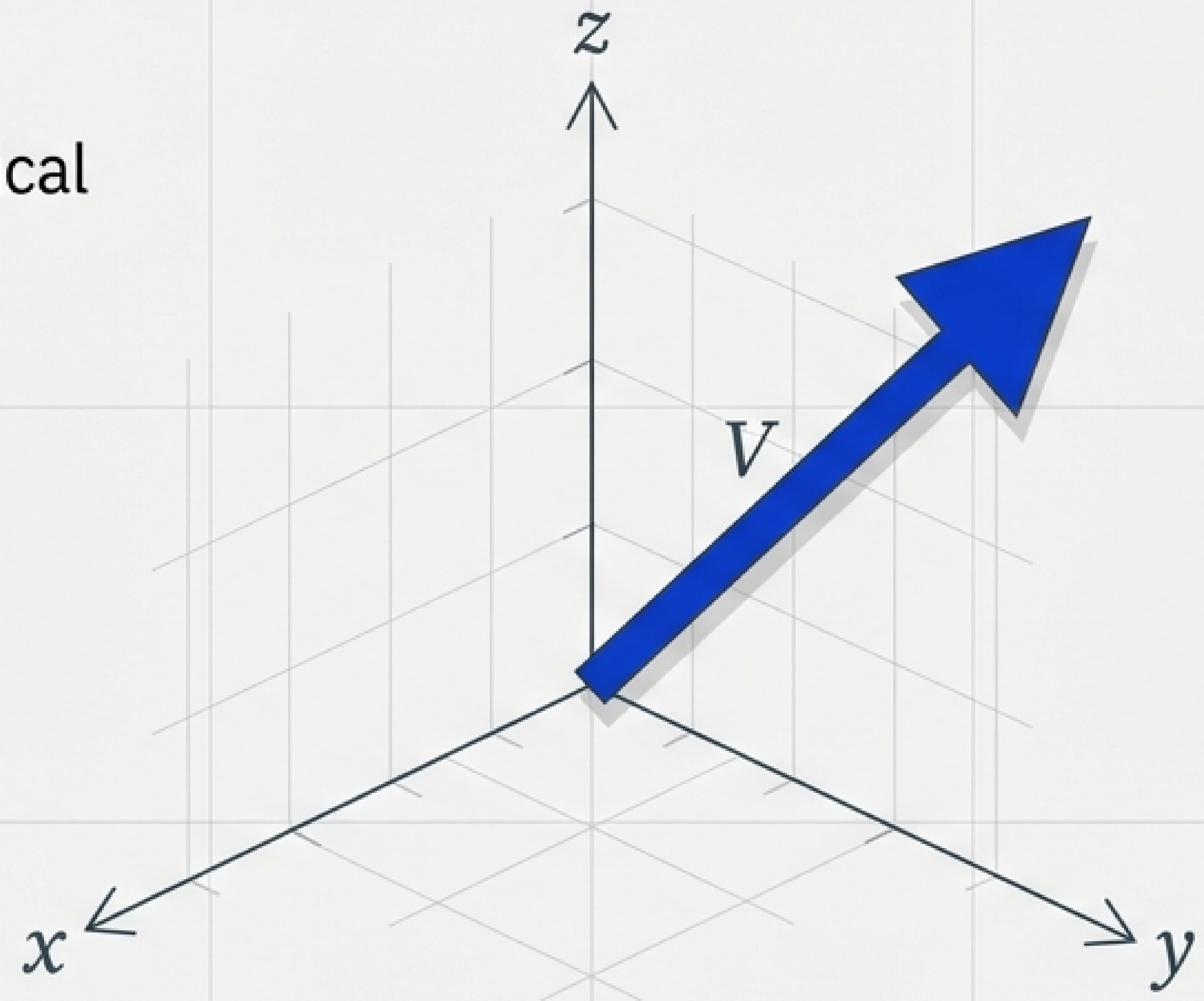
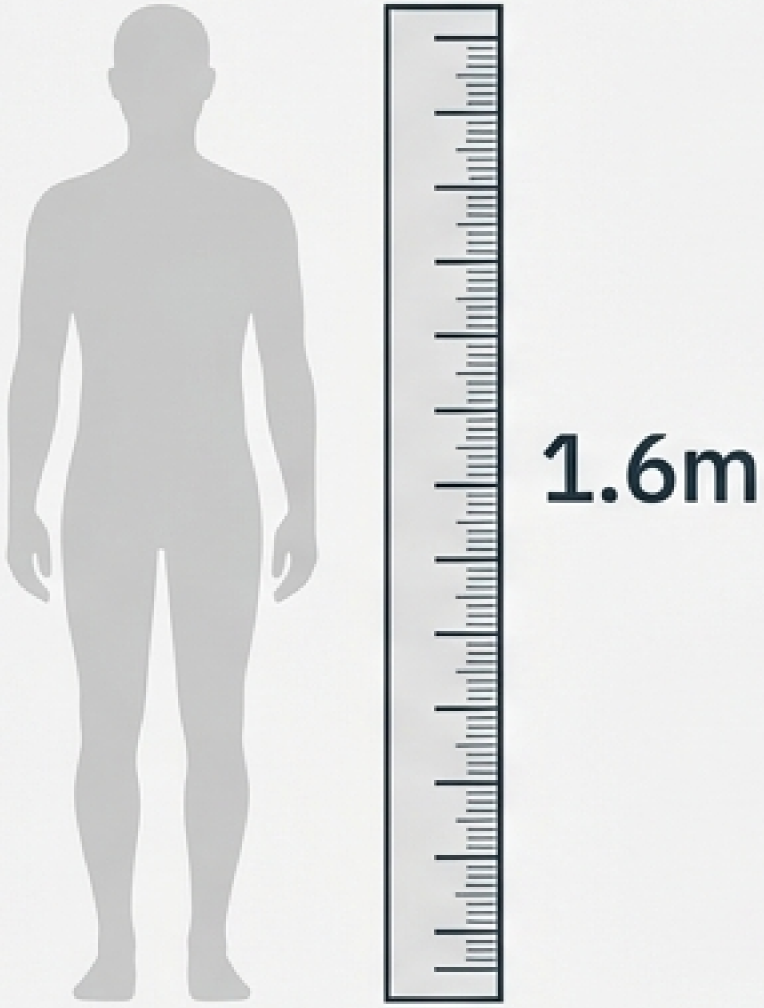
Vector Algebra: Navigating Magnitude & Direction

A toolkit for describing physical reality in 3D space.



Based on Chapter 10 Mathematics.

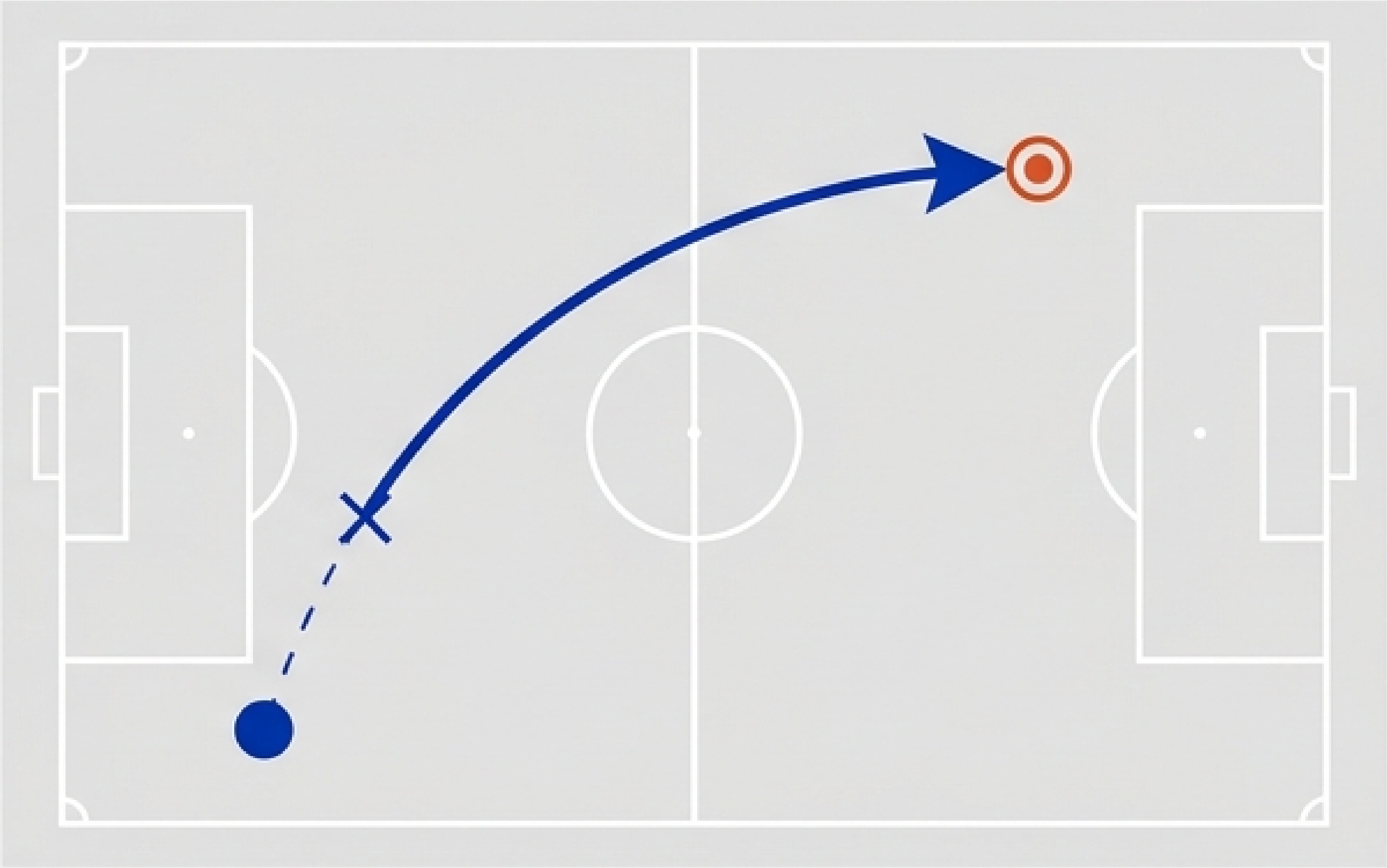
Scalar Quantities



Quantities involving only Magnitude.

- Length, Mass, Time, Volume, Speed, Temperature, Money

Vector Quantities

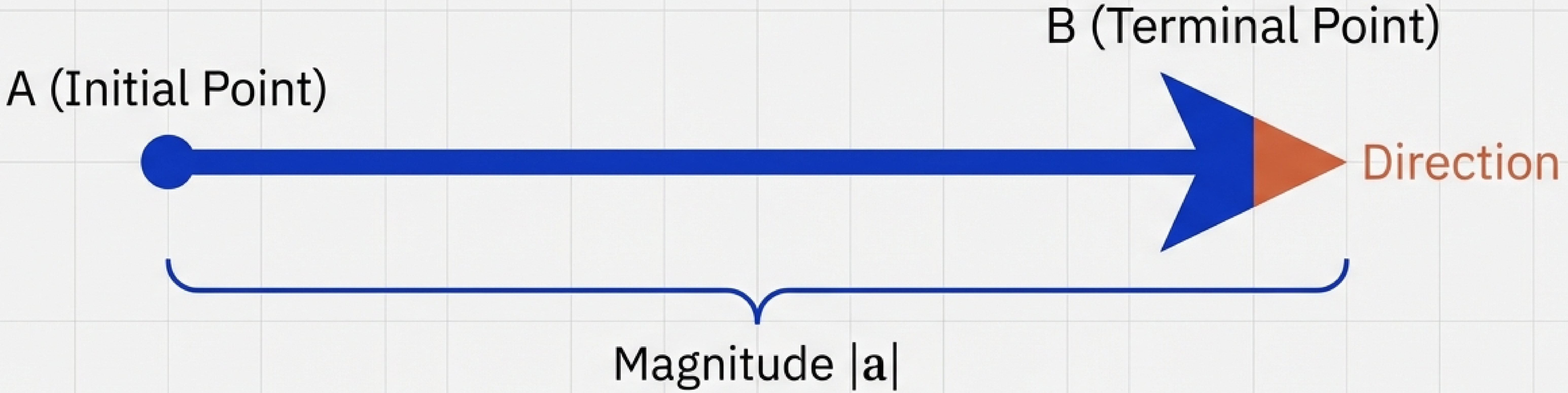


Quantities involving Magnitude AND Direction.

- Displacement, Velocity, Force, Acceleration, Momentum

Real numbers tell us ‘how much.” Vectors tell us ‘how much’ and ‘where.”

The Anatomy of a Directed Line Segment



Critical Rule:

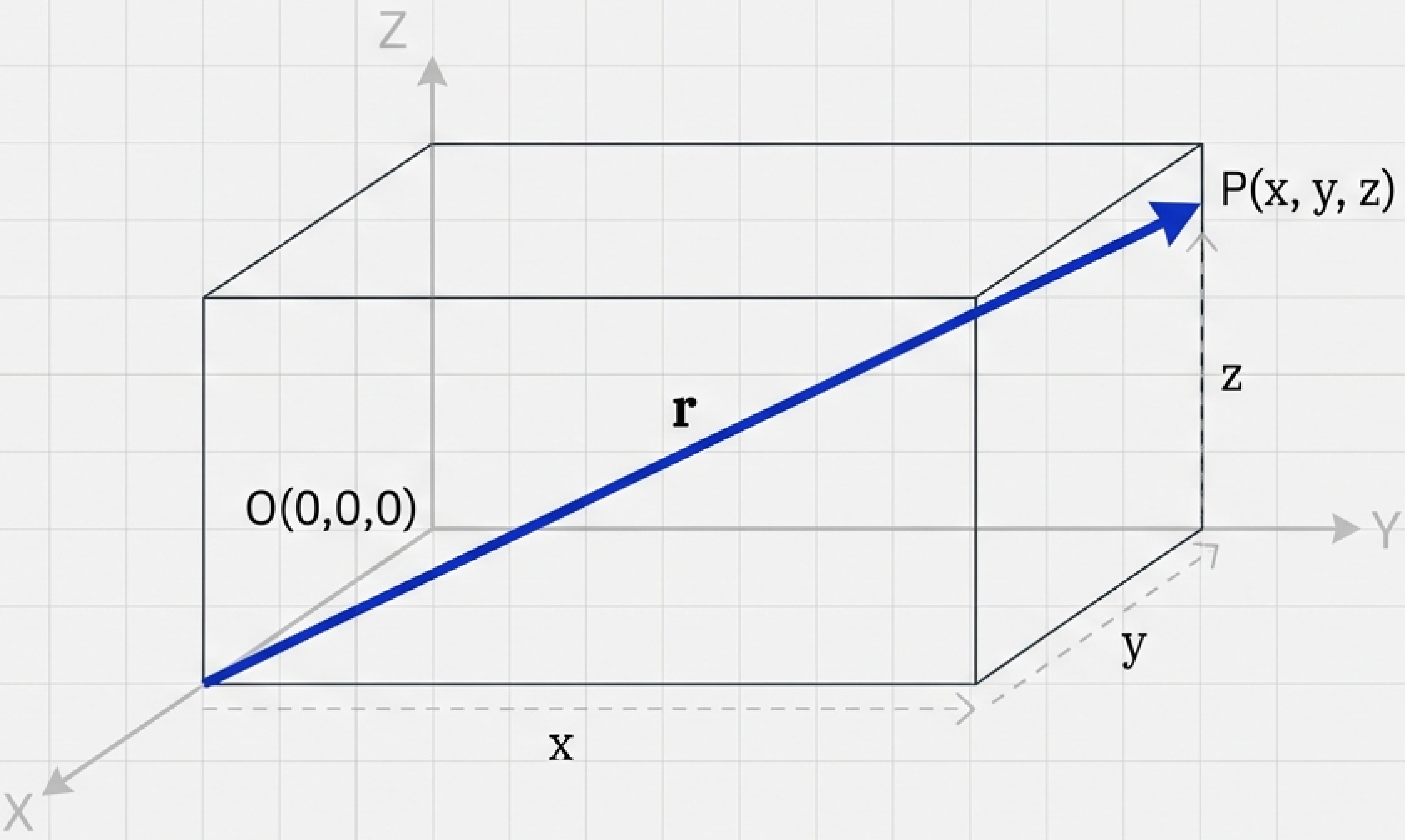
Since length is never negative, the notation $|a| < 0$ has no meaning.

Position Vectors in 3D Space

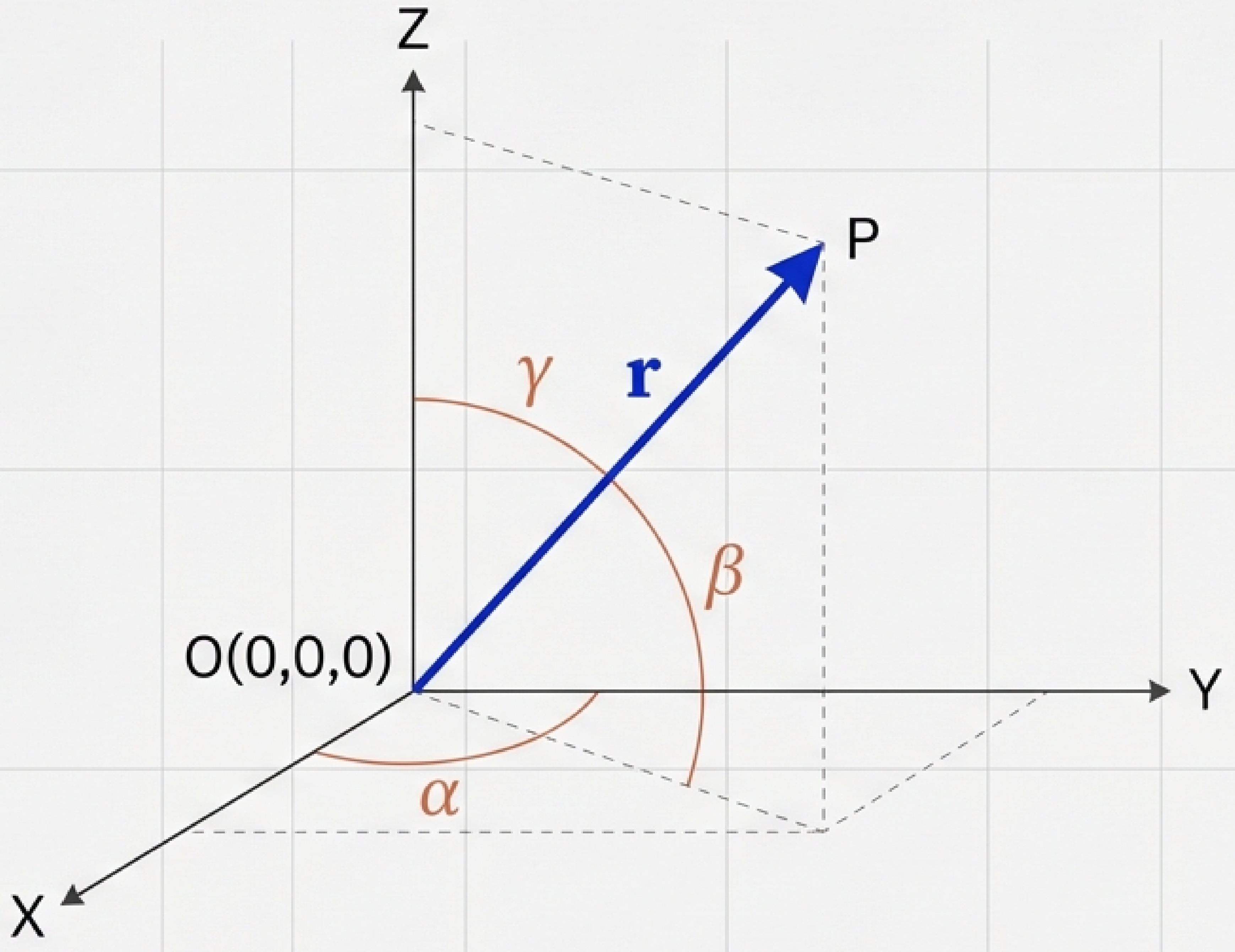
To locate point P(x, y, z), we use position vector $\mathbf{r}$.

Magnitude Formula:

$$|\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$$



Mapping Orientation: Direction Cosines



Direction Cosines:

$$l = \cos\alpha$$
$$m = \cos\beta$$
$$n = \cos\gamma$$

The Identity: $l^2 + m^2 + n^2 = 1$

The Vector Glossary

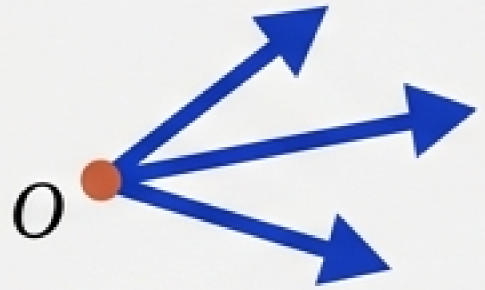
Zero Vector

Initial & terminal points coincide. Magnitude = 0.

Unit Vector

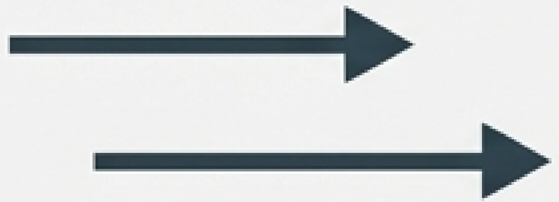
Magnitude is unity.

Coinitial Vectors



Same initial point.

Collinear Vectors



Parallel to the same line.

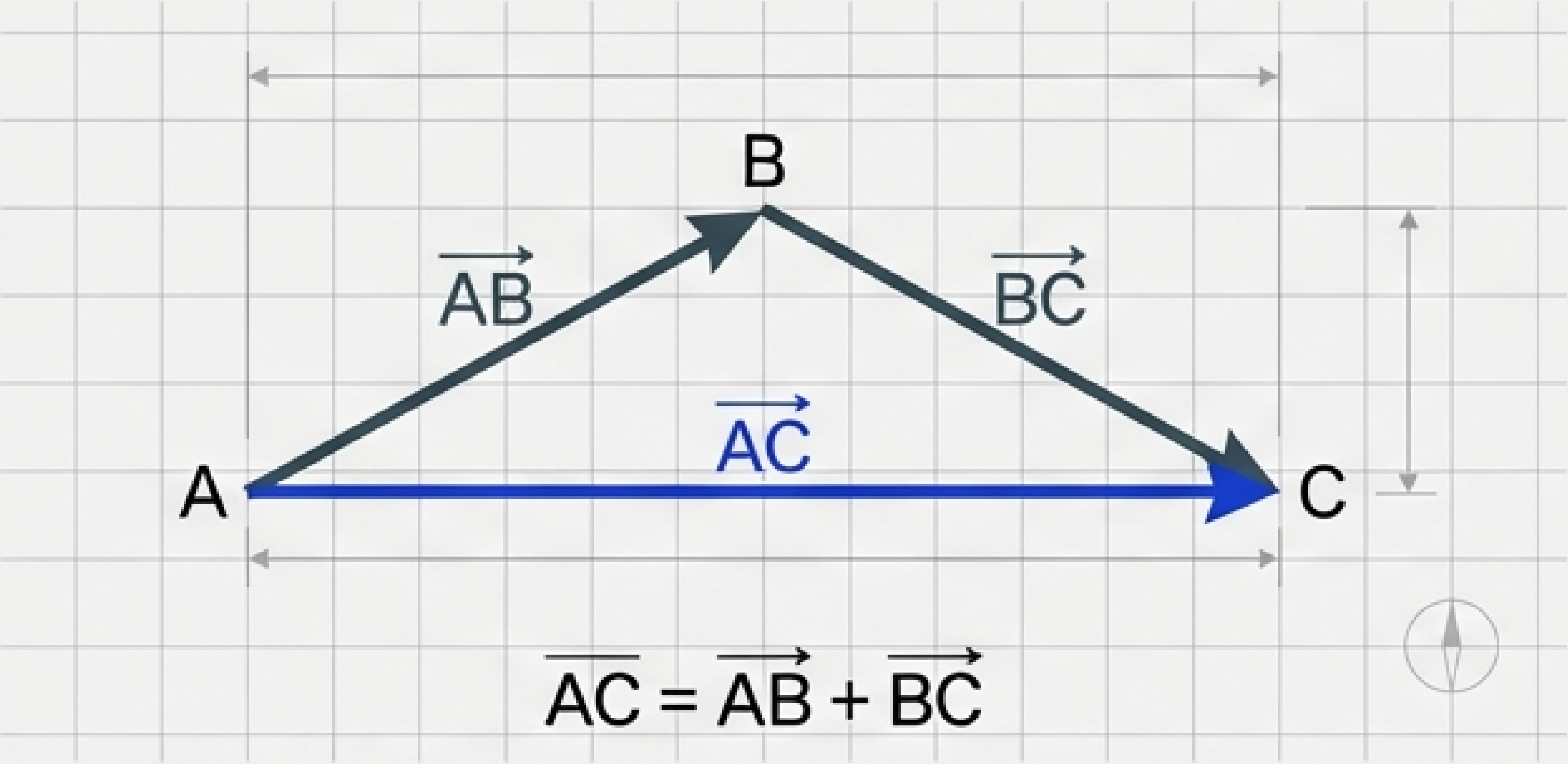
Equal Vectors

Same magnitude & direction.

Negative Vector

Same magnitude, opposite direction.

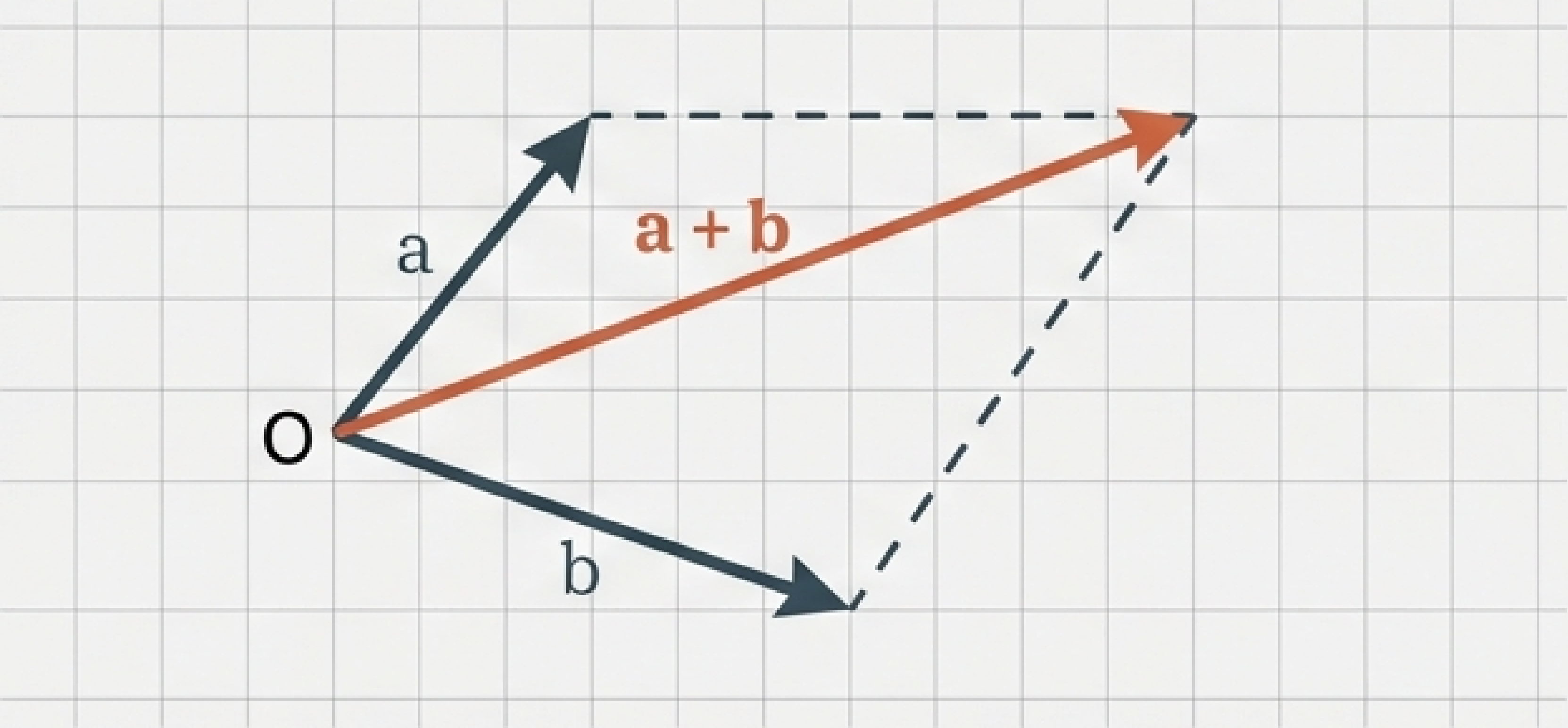
Vector Addition: The Geometry of Displacement



Triangle Law

IBM Plex Sans

AC = AB + BC



Parallelogram Law

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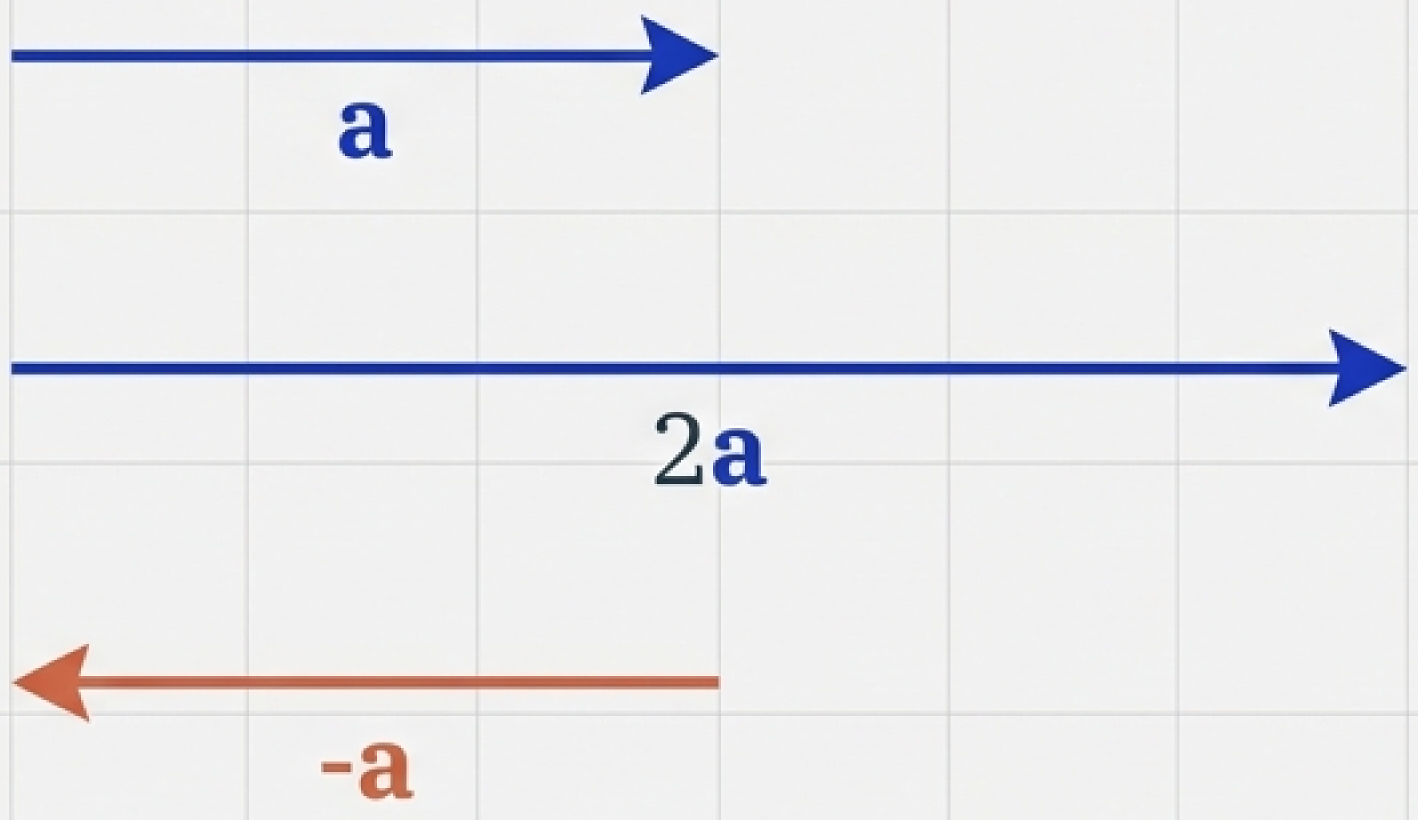
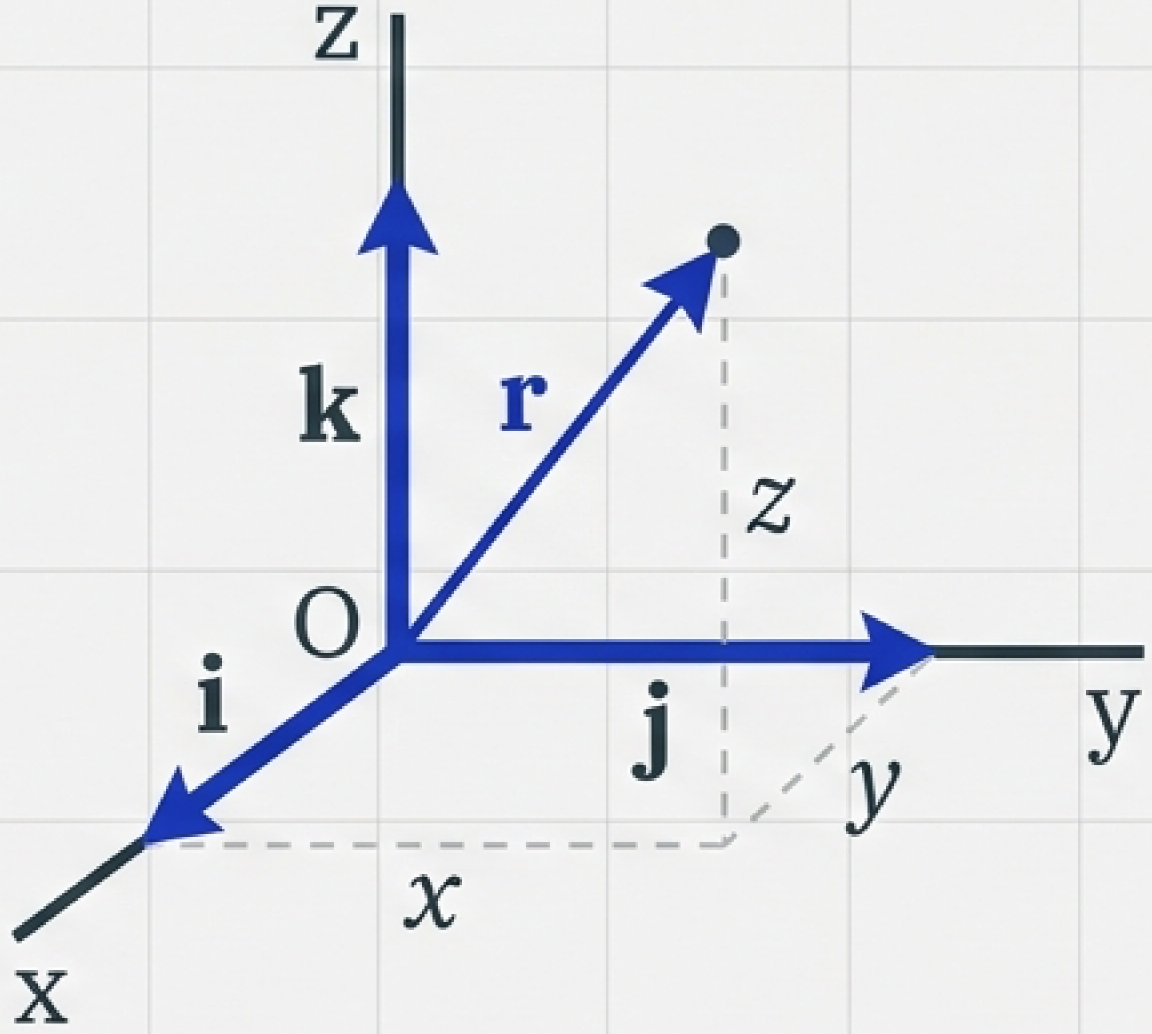
Useful for simultaneous forces.

Components & Scalar Multiplication

Scalar Multiplication



Component Form



$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

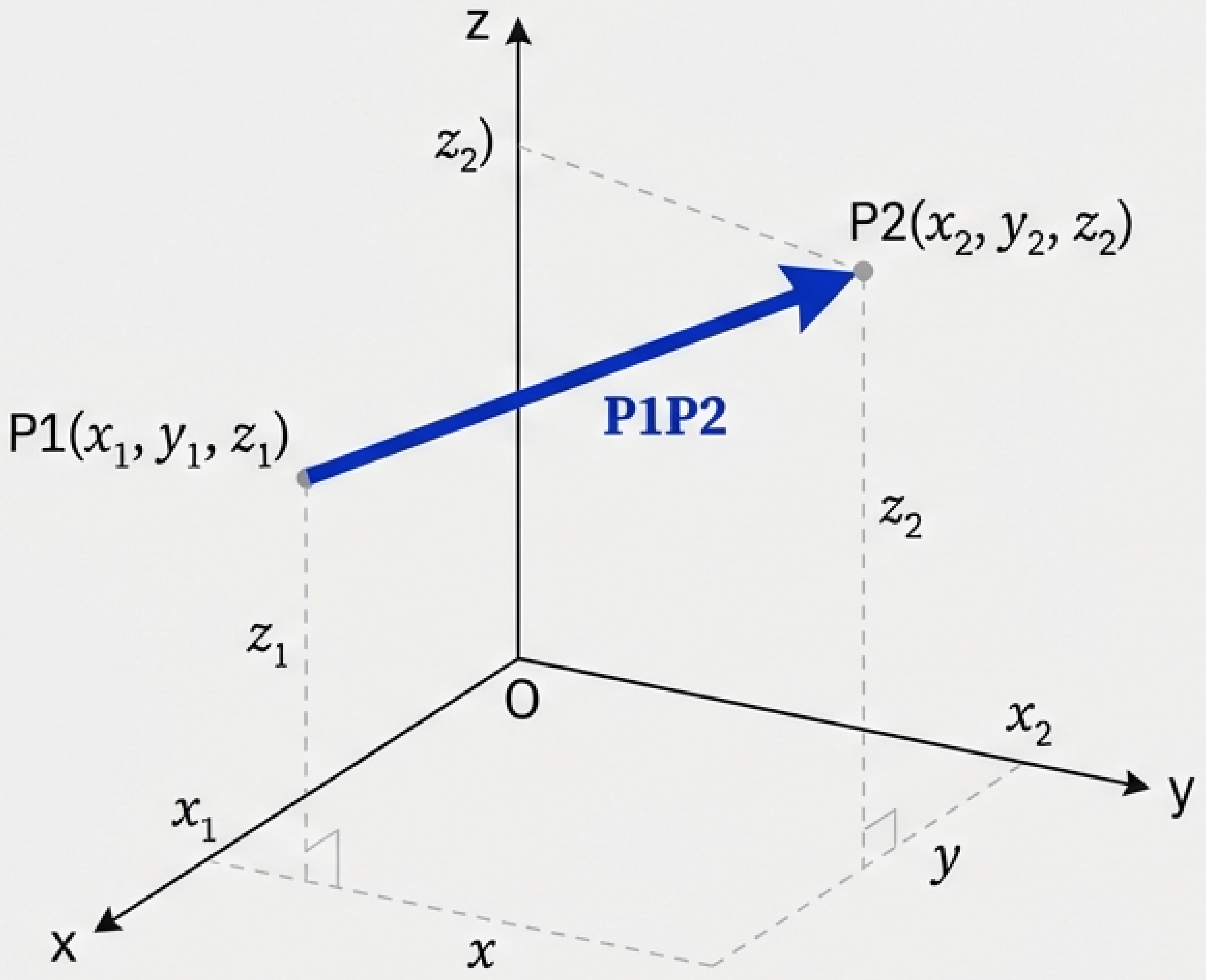
Vector Joining Two Points

To find the vector between two arbitrary points:

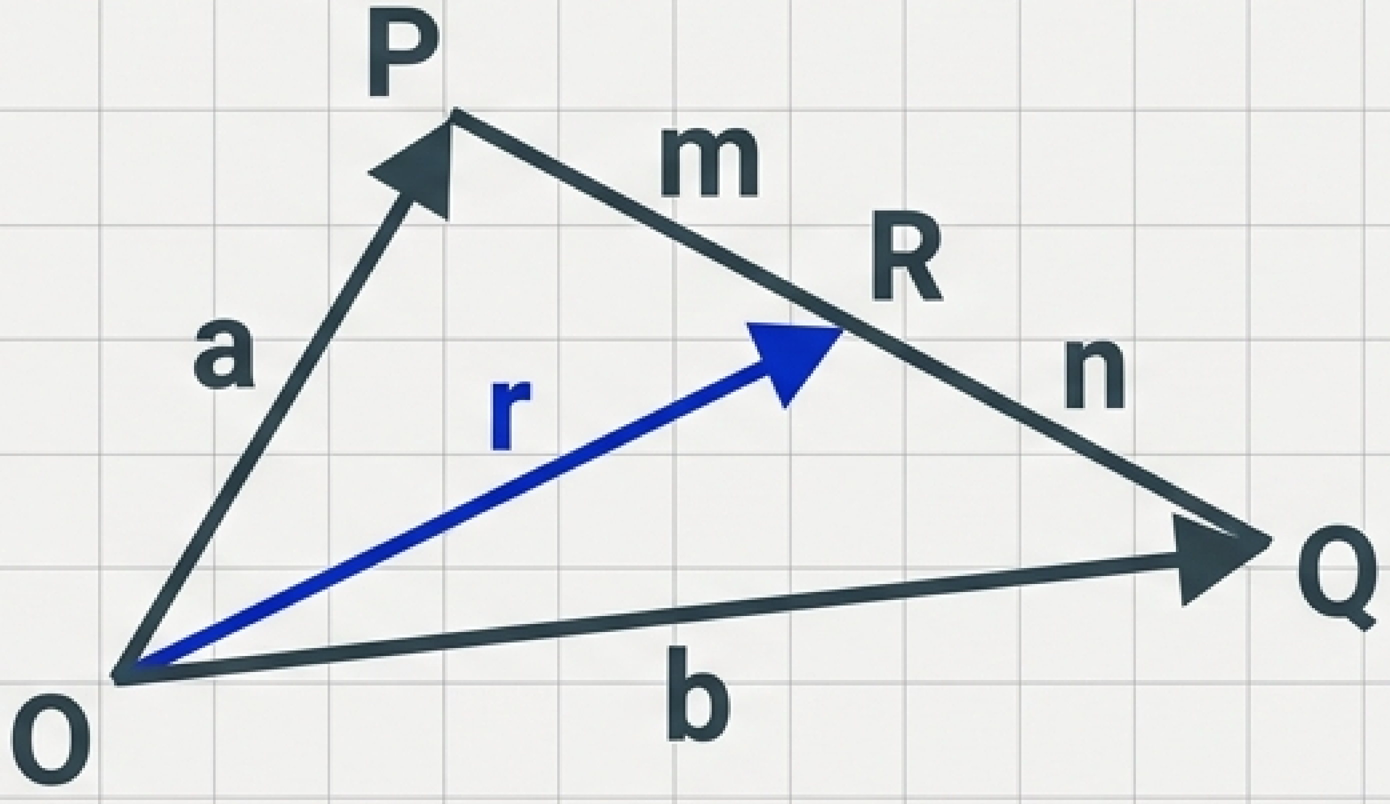
Formula: $\mathbf{P1P2} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$

Magnitude:

Formula: $|\mathbf{P1P2}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$



The Section Formula



Internal Division:

$$\mathbf{r} = \frac{mb + na}{m + n}$$

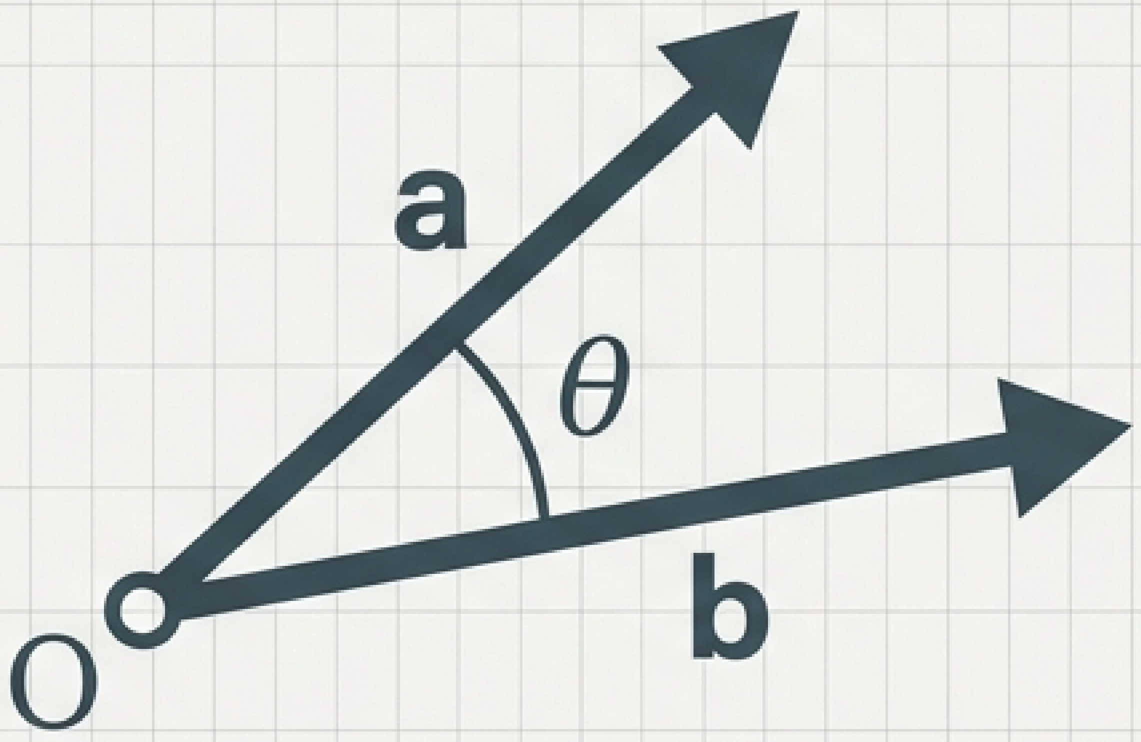
External Division:

$$\mathbf{r} = \frac{mb - na}{m - n}$$

Midpoint (m=n):

$$\mathbf{r} = \frac{\mathbf{a} + \mathbf{b}}{2}$$

The Scalar (Dot) Product



$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$$

- If $\mathbf{a} \cdot \mathbf{b} = 0$, the vectors are **PERPENDICULAR** (90°).
- If $\theta = 0$, $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|$ (**Maximum**).

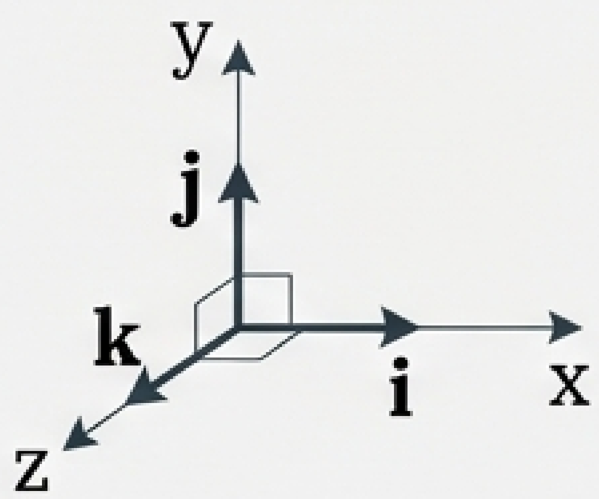
Calculating the Dot Product

The Logic

Unit vectors are perpendicular or parallel:

$$\mathbf{i} \cdot \mathbf{i} = 1$$

$$\mathbf{i} \cdot \mathbf{j} = 0$$



The Component Formula

If $\mathbf{a} = a1\mathbf{i} + a2\mathbf{j} + a3\mathbf{k}$
and $\mathbf{b} = b1\mathbf{i} + b2\mathbf{j} + b3\mathbf{k}$

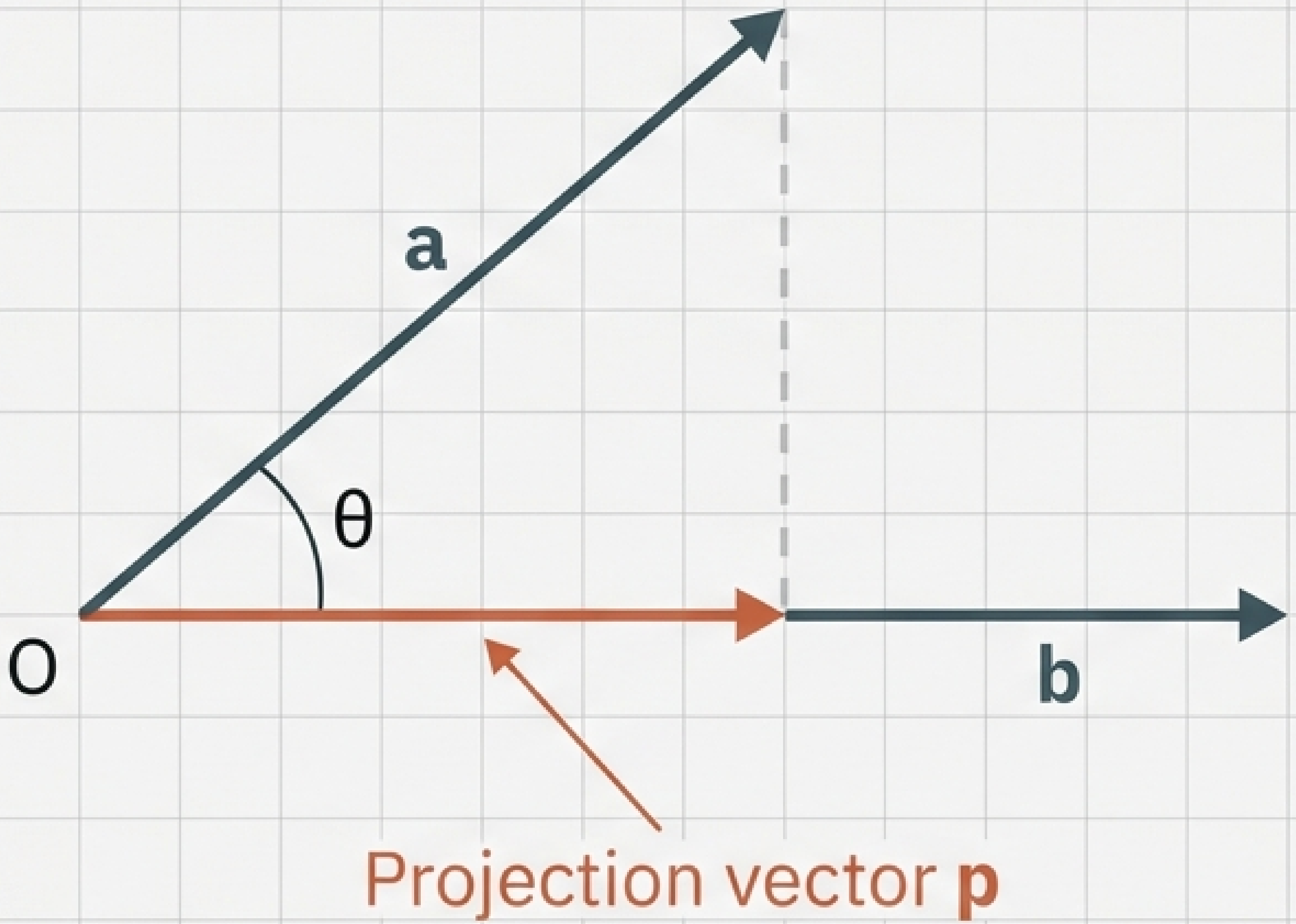
Then:

$$\boxed{\mathbf{a} \cdot \mathbf{b} = a1b1 + a2b2 + a3b3}$$

Finding the Angle:

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

Projection of a Vector



The “shadow” vector **a** casts on vector **b**.

$$\text{Projection of } \mathbf{a} \text{ on } \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$$

Critical Inequalities

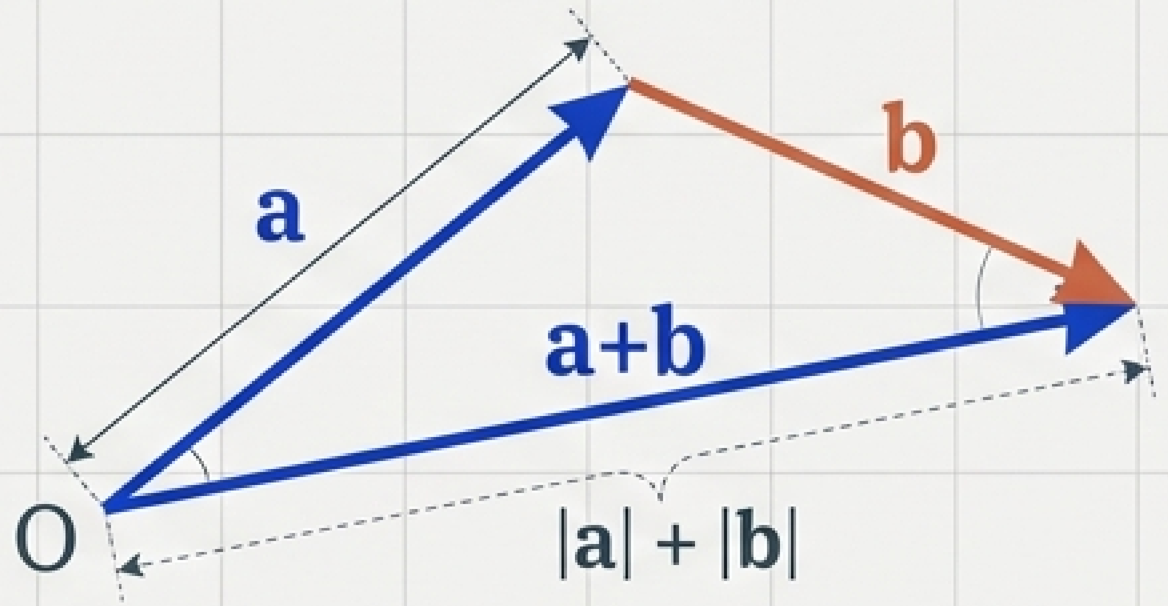
Cauchy-Schwartz Inequality

$$|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}|\ |\mathbf{b}|$$

The dot product never exceeds the product of magnitudes.

Triangle Inequality

$$|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$$



The straight path (resultant) is shorter than the two sides combined.

The Vector Toolkit: Key Formulas

Concept	Formula
Position & Magnitude	$\mathbf{r} = \mathbf{x}\mathbf{i} + y\mathbf{j} + \mathbf{z}\mathbf{k} \quad \lvert\mathbf{r}\rvert = \sqrt{(x^2 + y^2 + z^2)}$
Unit Vector	$\hat{\mathbf{a}} = \mathbf{a} \,/\, \lvert\mathbf{a}\rvert$
Vector Addition	$\mathbf{a} + \mathbf{b} = (a_1 + b_1)\mathbf{i} + (a_2 + b_2)\mathbf{j} + (a_3 + b_3)\mathbf{k}$
Dot Product	$\mathbf{a} \cdot \mathbf{b} = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta = a_1b_1 + a_2b_2 + a_3b_3$
Perpendicularity	$\mathbf{a} \cdot \mathbf{b} = 0$
Projection of **a** on **b**	$(\mathbf{a} \cdot \mathbf{b}) \,/\, \lvert\mathbf{b}\rvert$